

PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Smart Complaint Management System

Campus Facility Complaint Registration, Routing and Resolution Tracking

SYNOPSIS / PROJECT PROPOSAL

Item	Details
Group / Course	Team 6 / UE24CS341A — Software Engineering
Document version	2.0 — consolidated
Product scope	SCMS Release 1.0; six planned Agile sprints
Document status	**Prepared for submission**

Team profile

Member / ownership	SRN / PRN
Anushka K Kishore — F1	PES1UG24CS074 / PES1202400461
Atul Suresh — F2	PES1UG24CS098 / PES1202402181
Chaitanya Krishna Makkena — F3	PES1UG24CS122 / PES1202403729
Tanmay Sridhar — F4	PES1UG25CS849 / PES1202503400

1. Project description and problem

SCMS is a browser-based portal for campus and hostel facility complaints at PES University. It replaces a fragmented process of office registers, messages, phone calls, verbal reports and repeat visits. Without a shared record, complainants cannot reliably check acknowledgement, responsibility or progress, and staff lack a consolidated pending-work view.

A Complainant signs in, records a category, location and description, optionally uploads evidence, and receives a complaint ID. Configured rules route it to a department; the Department Head assigns Department Staff, who record progress and resolution. The Complainant tracks history, confirms resolution or reopens it. Anonymous intake uses a separate tracking reference without an account. Due dates, escalation and reports make overdue work visible.

1.1 Objectives

- Record every accepted complaint with a unique ID and responsible department.
- Provide authorized tracking, evidence and chronological history.
- Route by category/location and escalate missed resolution targets.
- Support anonymous reporting without identifying the reporter.
- Notify affected parties and collect confirmation/feedback.
- Produce department and campus reports with CSV export.
- Build and validate the system through four meaningful functional contributions.

1.2 Users and stakeholder

Actor	Purpose
Complainant	Student, hostel resident, faculty or non-teaching staff: submit/track own work and give feedback.
Anonymous Complainant	Submit without login; track and give permitted feedback using reference.
Department Staff	Resolver: work assigned complaints and record resolution.
Department Head	Warden where applicable: allocate work, handle escalations and monitor own department.
Administrator	Maintain users/master data/rules/SLA and campus reports.

Institutional stakeholder: Campus Facility and Hostel Administration Office. The manual-process description and exact master-data/SLA values require stakeholder confirmation.

2. Scope and functional features

Area	Planned functionality
User and Access Management	Registration, login/logout, password rules, temporary lockout, session expiry, four roles and department membership.
Complaint Management	Structured intake, active-value checks, unique ID, up to 3 JPEG/PNG/PDF files of at most 5 MB each, anonymous intake and restricted reference lookup, own-complaint tracking.
Assignment and Resolution	Active-rule routing and default queue; queue ordering; allocation/reallocation; priority/category due date; scheduled escalation; controlled lifecycle; history; resolution evidence; confirmation/reopen/auto-close; notices.
Administration and Monitoring	Configure users/roles/categories/sub-categories/locations/departments/rules/SLA; dashboards; grouped counts, resolution times, overdue work and scoped CSV.

2.1 Workflow and limits

Submit → automatic department routing → resolver assignment → progress → resolution → confirmation/feedback or reopen → close. Escalation is a separate flag/level, not a complaint state and not automatic reassignment. Anonymous reporting has no stored identity/email and no login requirement. Physical repair occurs outside SCMS.

Release 1.0 is single-campus, English-only and online. It excludes native mobile apps, SMS/WhatsApp, ERP/HRMS/payroll integration, financial transactions, repair inventory/costing, AI/ML classification or duplicate detection, IoT, live chat, offline operation, multi-campus tenancy and working-hour/holiday SLA calendars.

2.2 Security and quality

SO-1 confidentiality/privacy; SO-2 integrity/accountability; SO-3 operational continuity. SRS controls cover authentication, current role/object/file permissions, validation/escaping, CSRF, HTTPS, anonymous privacy, safe errors and logs. Performance and coverage numbers are acceptance targets.

3. Team ownership

Each member owns one end-to-end functional area, including screens, data handling, validation, tests and documentation. UI/database/testing-only duties do not replace functional ownership.

Area / Member	Owned functionality
F1 — Foundation and Access Anushka K Kishore	Registration, login/logout, password/lockout/session rules, RBAC, departments/staff membership, administrator screens/reports/CSV.
F2 — Complaint Intake and Evidence Atul Suresh	Complaint form/validation, unique ID, evidence upload/viewing, anonymous submission/reference tracking and specified lookup throttling.
F3 — Routing, Assignment and Escalation Chaitanya Krishna Makkena	Categories/SLA, rules, automatic routing, queues, resolver assignment/reassignment, due dates and scheduled escalation.
F4 — Resolution, Notifications and Tracking Tanmay Sridhar	State/history, progress/resolution, reject/reopen/auto-close, in-app/email notices, feedback/rating, Complainant/Head dashboards.

3.1 Handover between areas

F1 supplies users, roles, departments and the base UI. F2 supplies validated complaint records/evidence and references. F3 supplies department routing, resolver allocation, due dates and escalation. F4 completes resolution, notices, feedback, tracking and closure. Administrator reports/exports remain F1; Complainant/Head dashboards remain F4.

3.2 Shared responsibilities

Member	Shared responsibility
Anushka	Access-rule and security validation.
Atul	pytest fixtures, coverage reporting and demonstration data.
Chaitanya	Jenkins build/test/deployment workflow.
Tanmay	Repository/branch policy, GitHub Projects backlog and documentation upkeep.

All members participate in requirements/design discussion, integration, genuine commits, peer review and final explanation.

4. Work plan and technology

Sprint	Planned output / acceptance evidence
1	Setup, authentication/RBAC, repo/backlog, pytest and CI; working role login and genuine member commits/pipeline evidence when available.
2	Departments/admin, intake/ID/evidence; validated complaint and validation-unit tests.
3	Lockout/session, anonymous lookup, categories/rules, routing/queue/allocation; access and routing walkthrough.
4	Reassignment, due dates, lifecycle/history/rejection; submission-to-resolution demonstration and transition tests.
5	Escalation, notifications, reopen/auto-close/feedback; scheduled-event and feedback test evidence.
6	Dashboards/report/export, security, defects, coverage, final docs/demo; verified reports and real validation results.

The six-sprint plan uses Agile iterations to add visible complaint-workflow capabilities, incorporate feedback and perform regression testing before the next step.

4.1 Selected / required tools

Purpose	Selected tools
Application / UI	Python + Django; Django templates, HTML/CSS/JavaScript.
Data / files	PostgreSQL deployment, SQLite local; Django ORM and file storage.
Notifications / timing	Django email backend over institutional SMTP; scheduled Django management commands.
Development / validation	Git/GitHub, GitHub Projects, pytest/Django test client and coverage.
Course delivery obligations	Jenkins CI/CD, Docker, code review, security validation and static analysis (SonarQube or equivalent).